

ECONOMETRICS I

Problem Set 6: Instrumental variables

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1. CRIME AND POLICE: SIMULTANEITY AND IV (SIMULATION)

Consider the following simultaneous equations model that describes the relationship between crime rates and police deployment:

$$\text{Crime} = \alpha_c + \beta_c \text{Police} + \gamma_c \text{Unemp} + \varepsilon_c \quad (\text{eq.C})$$

$$\text{Police} = \alpha_p + \beta_p \text{Crime} + \gamma_p \text{RandBudg} + \varepsilon_p \quad (\text{eq.P})$$

where:

- **Crime** is the crime rate.
- **Police** is police presence.
- **Unemp** is the unemployment rate.
- **RandBudg** is a randomly assigned police budget (think of the government running an experiment).
- ε_c and ε_p are exogenous error terms.

Run the provided R code **crime police** to generate the data and estimate the model. In the code, the true parameter values are: $\beta_c = -2$ (effect of police on crime) and $\beta_p = 1.5$ (effect of crime on police).

- (a) You want to estimate the effect of Police on Crime. Plot Crime vs. Police and explain why ordinary least squares (OLS) regression of Crime on Police would yield biased estimates of β_c .

Solution: When plotting Crime vs. Police, we observe a positive relationship in the scatter plot. This seems counterintuitive if police reduce crime ($\beta_c = -2$), but the positive correlation occurs due to simultaneity bias.

OLS would be biased because:

- Reverse causality: Police reduces Crime, but high-crime cities deploy more police ($\beta_p = 1.5 > 0$). The OLS estimate mixes these effects—it captures how police affect crime *and* how crime affects police deployment, and more (see below).
- The problem can be understood as a violation of the assumption $\mathbb{E}(u|X) = 0$ or $\text{cov}(u, x) = 0$ in the regression

$$\text{Crime} = \beta_1 + \beta_2 \text{Police} + u.$$

To see why, note that subtracting (eq.C) from (eq.P) and solving for Crime yields:

$$\text{Crime} = \underbrace{\frac{\alpha_c + \alpha_p}{1 + \beta_p}}_{\beta_1} + \underbrace{\frac{1 + \beta_c}{1 + \beta_p}}_{\beta_2} \text{Police} + \underbrace{\frac{\gamma_c \text{Unemp} - \gamma_p \text{RandBudg}}{1 + \beta_p}}_u + \frac{\varepsilon_c - \varepsilon_p}{1 + \beta_p}$$

Here we have two problems. First, β_2 different from the parameter of interest (β_c). Second, the regression would not even correctly estimate $\frac{1+\beta_c}{1+\beta_p}$. This is because the error term u contains two sources of variance correlated with Police: RandBudg and ε_p . (Note that even controlling RandBudg and Unemp would not retrieve the $\frac{1+\beta_c}{1+\beta_p}$, since the unobserved ε_p is also correlated with Police.)

- Without addressing simultaneity, we cannot identify the causal effect β_c

- (b) Given the equations above, what variable could be used as an instrument to estimate the effect of Police on Crime? Discuss the relevance and exclusion restrictions.

Solution: To estimate the effect of Police on Crime (β_c), we can use **Rand-Budg** as an instrument.

Relevance condition:

- RandBudg must be correlated with Police
- In the Police equation: $\text{Police} = \alpha_p + \beta_p \text{Crime} + \gamma_p \text{RandBudg} + \varepsilon_p$
- Since $\gamma_p \neq 0$, budget allocations directly affect police deployment
- First stage regression shows RandBudg significantly predicts Police
- More budget allows hiring more police officers

Exclusion restriction:

- Random assignment $\implies$ natural experiment.
- By assumption (see equations), RandBudg affects Crime only through its

effect on Police. Plausible real-life violation of the exclusion restriction: if police forces hired locally, affecting local unemployment.

- In our simulations: exclusion restriction holds.

- (c) The manual 2SLS approach involves two stages. Write the equations for both stages and interpret the results from the R output.

Solution: First stage: $\text{Police} \sim \text{RandBudg} + \text{Unemp}$

- Estimates: $\widehat{\text{Police}} = \hat{\delta}_0 + \hat{\delta}_1 \text{RandBudg} + \hat{\delta}_2 \text{Unemp}$
- Creates predicted police presence using only exogenous variation from RandBudg and Unemp

Second stage: $\text{Crime} \sim \widehat{\text{Police}} + \text{Unemp}$

- Estimates: $\widehat{\text{Crime}} = \hat{\alpha}_c + \hat{\beta}_c \widehat{\text{Police}} + \hat{\gamma}_c \text{Unemp}$
- Uses only the exogenous variation in police from the first stage
- The coefficient $\hat{\beta}_c$ is the IV estimate of police effect on crime

From R output: The 2SLS estimate should recover approximately -2 , showing that more police reduce crime.

- (d) Given the equations above, what variable could be used as an instrument to estimate the effect of Crime on Police? Discuss the relevance and exclusion restrictions.

Solution: To estimate the effect of Crime on Police (β_p), we can use **Unemp** as an instrument.

Relevance condition:

- Unemp must be correlated with Crime
- In the Crime equation: $\text{Crime} = \alpha_c + \beta_c \text{Police} + \gamma_c \text{Unemp} + \varepsilon_c$
- Since $\gamma_c \neq 0$, unemployment affects crime rates
- First stage regression shows Unemp significantly predicts Crime

Exclusion restriction:

- Unemp should affect Police only through its effect on Crime
- In our simulation, we know already: Unemployment affects police deployment only via its impact on crime rates
- In reality, we should think about possible issues. For example, be sure

that unemployment doesn't affect police hiring decisions (e.g., that cities don't hire more police just because unemployment is high, or that economic conditions that relate with unemployment don't determine the amount of police)

- (e) Compare the 2SLS estimates with the true parameter values.

Solution: Comparison:

- Police effect on Crime: Estimate ≈ -2 vs True $= -2$
- Crime effect on Police: Estimate ≈ 1.5 vs True $= 1.5$

This is no surprise: because we created the data, we knew already which variables are exogenous in the system. This is of course much more difficult in practice. You have to think hard about what may cause what in the real world. Finding good instruments is a challenge.

2. DISTANCE TO COLLEGE AND RETURN TO EDUCATION (SIMULATION)

Consider the simulation code given in **collegeDistanceSimulation**. It generates the following (artificial) data:

1. Observable variables:

- **educ**: a person's years of education.
- **lwage**: the person's log wage after going to college.
- **urban**: indicates with 1 if the person was raised in an urban center; 0 if raised in a rural place.
- **ldistance**: the log distance to the closest college relative to the living location prior to study.

2. Unobservable processes and variables

- **talent**: represents the talent of a person. It is unobservable but it a variable determining a person's wage.
- **Data generating process (DGP):**

```
talent <- runif(n_obs,0,10) # Person's talent (unobserved)
urban <- rbinom(n=n_obs, size=1, prob=0.5) # Urban dummy

ldistance <- 1 -urban + rnorm(n_obs) # Log college distance
educ <- 6-gamma*ldistance + talent + rnorm(n_obs) # Years of educ.
educ <- round(educ ,0) # drop decimals
```

```

beta_true <- .05 # unknown parameter to be estimated
# log wages:
lwage <- beta_true*educ + 0.5*urban + .02*talent + rnorm(n_obs)

```

Run the code with `n_obs <- 100000` and `gamma <- 1` (seed 1234). You should get the results of table 1.

Table 1: Results with $n = 100,000$ and $\gamma = 1$

	<i>Dependent variable: lwage</i>			
	(1)	(2)	(3)	(4)
	OLS	OLS	IV	IV
educ	0.079*** (0.001)	0.068*** (0.001)	0.151*** (0.003)	0.049*** (0.003)
urban		0.485*** (0.006)		0.503*** (0.007)
Constant	0.039*** (0.011)	-0.080*** (0.011)	-0.713*** (0.032)	0.105*** (0.032)
Observations	100,000	100,000	100,000	100,000
R ²	0.059	0.110	0.011	0.107
Adjusted R ²	0.059	0.110	0.011	0.107

Note:

*p<0.1; **p<0.05; ***p<0.01

- (a) Without yet considering the results of the table, why would distance to the nearest college serve in the real world as an instrument for education? Discuss the relevance and exclusion restrictions.

Solution: When evaluating an instrument, always start by thinking about the relevance and exclusion restrictions.

1. Relevance condition:

- Distance affects education decisions: people closer to colleges get more education, because of lower transportation costs, better information for those near colleges, known contacts, etc.
- This condition *can be tested*: First stage should show significant negative relationship.

2. Exclusion restriction:

- Reflects relevance ($\text{cov}[z, x] \neq 0$) and exogeneity ($\text{cov}[z, u] = 0$): Distance should affect wages *only* through education, not through other channels.
- Potential threats to exclusion:
 - Urban areas have both colleges and higher wages (the case in our simulation)
 - Family background correlates with location choices (not modeled in our simulation, but plausibly an issue in the real world).
- This condition *cannot be tested* empirically. In practice, a researcher needs to think hard about whether the condition holds or not. Researchers relying on instrumental variables often struggle to convince their peers about the exclusion restriction.

- (b) Now look at the table. You know that the actual coefficient of educ in the DGP is 5%. Why do columns (1) and (2) estimate a much larger value?

Solution: There is an upward bias through the omission of both talent and urban (both of which feature in the last line of the DGP). Recall from lecture 6 that the bias is given by $\beta_{\text{omitted}} \times \frac{\text{cov}(\text{included}, \text{omitted})}{\text{var}(\text{included})}$. In (2) the omitted variable is talent, which has a positive impact on lwage ($\beta_{\text{talent}} = 0.02$ in the DGP) and is positively correlated with educ (see educ in the DGP). The bias is even larger in (1) because also urban is omitted, which also has a positive effect on lwage ($\beta_{\text{urban}} = 0.5$ in the DGP) and is positively correlated with educ (through ldistance, which affects educ in the DGP).

- (c) The bias is even larger in (3). Why? Compare to (4)

Solution: In (3), we are instrumentalizing the years of education through the ldistance. While this may appear a good idea, the exercise is invalidated through a bias arising from the omission of urban. Urban areas generally offer higher wages. Yet in this specification we are omitting this direct effect of urban on wages, artificially inflating the effect that educ (which also correlates with urban, through ldistance) has on lwage.

In (4), by contrast, we are estimating a regression without omitted variables and the instrument does a very good job approximating the true coefficient value.

- (d) Rerun the code, now with a smaller values of γ . First try `gamma <- .01` and record your results. Then try `gamma <- .1`. What can you conclude?

Solution:

- With $\gamma = .01$, we get $\hat{\beta}_{IV} = -0.751$ ($se = 8.583$). This estimate is extremely far away from the true coefficient. Why? Because the first stage failed: `ldistance` has a t-statistic of just -0.103 , showing that the instrument is not sufficiently relevant.
- With $\gamma = .1$, we get $\hat{\beta}_{IV} = 0.041$ ($se = 0.036$). This estimate is much closer to the true coefficient. Now `ldistance` has a t-statistic of -9.084 in the first stage, which is highly significant: The instrument is relevant. Yet we see that even very significant first stages may not be sufficient to ensure a low variance of the second stage estimate. That is why, seeking efficiency, instrumental variable regression are often estimated with extremely large datasets.

- (e) OPTIONAL (because your computer may crash): rerun the whole code keeping the weak instrument (`gamma <- .1`) but with ten million observations (`n_obs <- 10000000`). Now column (4) should again produce an estimate close to 5%.

Solution: Your obtained coefficient of `educ` should this time be: `0.069***`, `0.068***`, `1.049***`, `0.049***` ($se = 0.003$)

This illustrates how a very large n leads to right estimates with instrumental variables. The IV estimator is indeed consistent. But, as with OLS estimates, which requires condition $cov(x, u) = 0$ to hold, consistency requires $cov(z, u) = 0$.

- (f) Increase the relevance of the instruments by setting $\gamma = 1$, but reduce the number of observations to $n = 1000$. What happened to the estimate in (4)?

Solution: Again, the instrument continues to be relevant (first stage $t = -10.737$) and the estimate is numerically accurate (0.050), but not sufficiently efficient to deliver a coefficient that is statistically different from zero. This highlights once more how IV estimations depend on a high number of observations.